



Incident report analysis

Applying the NIST framework to different situations encountered:

Summary	Recently, we became aware of an attack regarding our internal network which was taken down for 2 hours. This was found to be because of an overflow of ICMP packets, which led our network services to be unable to function. We blocked all incoming ICMP packets to temporarily halt the attack and restore critical network services. Upon further investigation, this DoS attack was possible because of our unconfigured firewall. We implemented rules to our firewall to limit the rate of incoming ICMP packets and prevent IP spoofing. We also added network monitoring software and an IDS/IPS system to filter our suspicious data packets.
Identify	ICMP flood attack was used. Our servers were affected due to the limited bandwidth caused by the attack.
Protect	- A new firewall rule to limit the rate of incoming ICMP packets (consider a NGFW) - An IDS/IPS system to filter out suspicious data packets
Detect	- Full packet transfer devices - Network monitoring software - IDS - Source IP address verification on the firewall to check for IP spoofing - Use MFA - Implement Chronicle to view user data - Keep a list of authorized users
Respond	- Contain incidents using network segmentation and VMs

	- NGFW, IPS - Log data can be used and analyzed with SIEM tools
Recover	- Need to know the cause of the attack, and that it is neutralized. - Regular data backups - Ensuring configurations afterwards match the baseline image

Reflections/Notes: In the future, external ICMP flood attacks can be blocked at the firewall. Then, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online.

EX:  Incident report analysis exemplar